



AI-Powered Auto Claims Handling

Faster, fairer claims — from First Notice of Loss through AI extraction, damage assessment, fraud scoring, and a settlement recommendation, with an adjuster making the final decision.

Quantum Shift AI Practitioner+ · Capstone Project (Batch #1)
Built with Claude · Cursor · GitHub Copilot

Group 7

TEAM MEMBERS

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Context and Why Now

Challenges and Impact

- **Business problem:** traditional claims handling is slow, labor-intensive, and inconsistent.
- **Challenges:** manual FNOL intake, triage, and adjudication create long cycle times; inconsistent decisions drive **claims leakage** and missed fraud.
- **Impact:** customers wait days/weeks → churn; adjusters stuck on routine claims; leakage & fraud erode the loss ratio.

Strategic Imperative — why now

- Rising claim volumes and **severity inflation** strain capacity.
- Customers expect **instant, digital-first** resolution.
- Carriers must scale lean and modernize core claims operations.

Shift in Operating Model

- AI moves from back-office analytics to the **front line of the claims workflow**.
- It reads submissions, assesses damage, scores fraud, and recommends decisions **in seconds**.
- Humans shift to **handling exceptions** and making the final call.

Current Challenges and Gaps

Existing process (manual)

- Customer calls/emails FNOL → rep keys in data.
- Claim sits in a queue for manual triage.
- Adjuster reads notes, requests photos, estimates damage by hand.
- Fraud caught only if an adjuster happens to notice.
- Decisions & settlements vary by individual.

Pain points & gaps

- **Manual activities:** re-keying data, reading free text, chasing missing info.
- **Productivity:** trivial and complex claims share one queue.
- **Inconsistency:** no standard severity / fraud / settlement logic.
- **Fraud control:** reactive and person-dependent.
- **Technology gap:** systems of record, not systems of judgment.
- **Business limitation:** can't scale volume without headcount.

Vision and Future State

What ClaimX enables

- ⚡ **Faster outcomes** — instant AI triage at submission.
- 🎯 **Consistent decisions** — every claim scored the same way.
- 😊 **Better experience** — real-time status, quick resolution.
- 💰 **Less effort & leakage** — evidence-based settlement.

Strategic outcomes

- Lower cost-per-claim and reduced cycle time.
- Improved loss ratio via consistency + fraud control.
- Adjuster capacity redirected to complex, high-value claims.

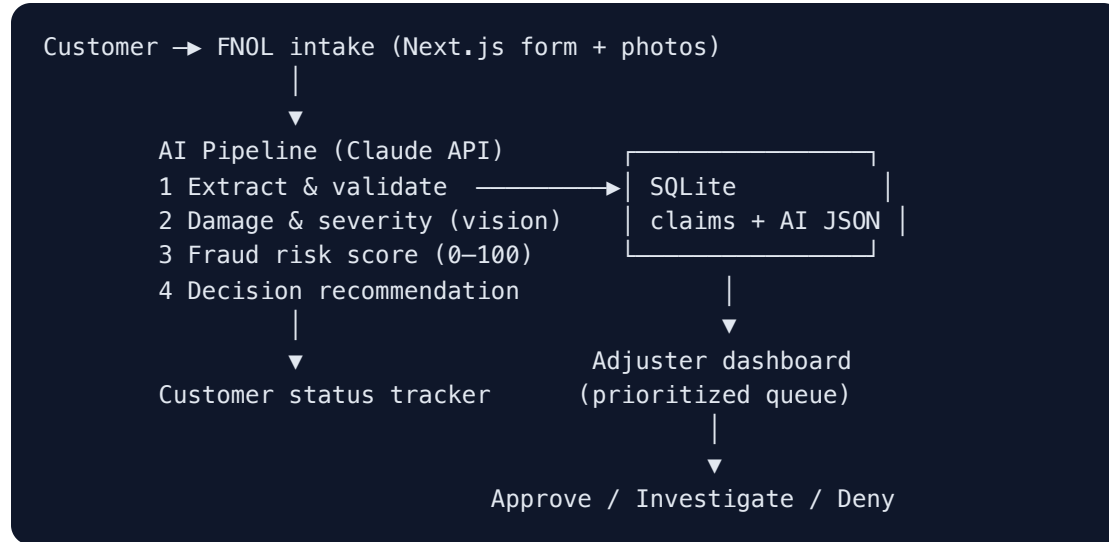
Future-state operating model

- **People:** adjusters become exception-handlers and decision-makers.
- **Process:** straight-through fast-tracking for clean claims; auto-routing of risky/complex ones.
- **AI:** always-on first reviewer — extract → assess → score → recommend, with a human-in-the-loop.

Future extensions: customer chatbot, automated notifications, analytics dashboards, multi-line support (home, health).

Solution Overview

Architecture / workflow



AI tools used

- **In product:** Claude API (claude-opus-4-8, vision + forced tool-use).
- **In development:** Claude Code / Cursor / GitHub Copilot.

Key features & capabilities

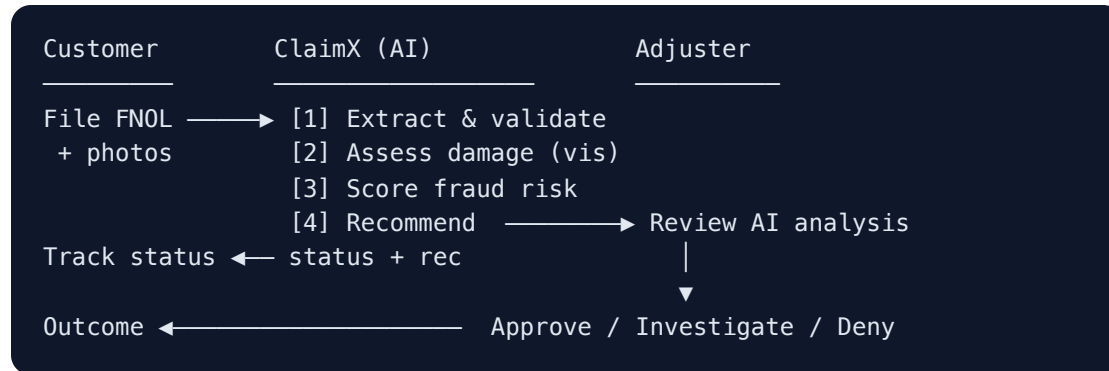
- Guided FNOL intake with photo upload and instant AI review.
- 4-step Claude pipeline → structured, typed JSON per claim.
- Adjuster dashboard auto-prioritized by fraud risk & status.
- One-click human decision; full customer status tracking.

Capability descriptions

- **Extraction** — normalize text/fields, flag gaps.
- **Damage** — vision over photos → severity + cost range.
- **Fraud** — 0-100 score with named red flags.
- **Recommendation** — disposition + settlement + rationale.

Business Flow and Role of AI

End-to-end workflow



User interactions

- Customer: submit claim, upload photos, track status & outcome.
- Adjuster: triage queue, review AI evidence, make the call.

AI intervention points (4)

- Intake — extraction & validation
- Damage — severity & estimate (vision)
- Fraud — risk scoring with red flags
- Decision — recommendation + settlement

Decision support & automation

- **Decision support:** AI gives recommendation, settlement, confidence, rationale — **the adjuster decides**.
- **Automation:** straight-through processing for low-risk claims; auto-routing of high-fraud/complex claims; future auto-drafted comms & SIU referrals.